

CALEB COOPER

SKILLS

- **Artificial Intelligence:** Prompt Engineering, AI Workflow Automation, Multi-Step Agent & Bot Development, Model Evaluation & Tuning, Chatbot Integration, Voice Agents, Agentic Systems, Rapid Prototyping
- **Programming:** Python, SQL, C#, ASP.NET, LangChain, PyTorch, NumPy, Pandas, Scikit-Learn, SciPy, Matplotlib, Seaborn
- **Data Engineering & Analytics:** ETL Pipelines, Data Quality Assurance, Microsoft Fabric, Apache Spark, Power BI, Azure, Tableau, Advanced Excel
- **Molecular Biotechnology:** PCR, Gel Electrophoresis, SDS-PAGE, HPLC, FPLC, Bioreactor Operations, DNA/ RNA Manipulation

EXPERIENCE

AI Developer – LLM Applications (2024 – Present)

Clients: Outlier AI, Snorkel AI, Telus AI

- **Model Development:** Fine – tune and adapt large language models (GPT-4), Claude, Gemini) for real-world use cases, blending prompt engineering with lightweight Python scripting and data driven evaluation.
- **Evaluation & Tuning:** Design structured test suites – coding, math, and domain specific prompts – to measure model accuracy, reliability, and safety. Feed findings into iterative tuning cycles to improve overall performance.
- **Prompt & Workflow Design:** Experiment with templates, few-shot setups, retrieval steps, and chain-of-thought prompts to make outputs more stable and reduce error rates in practical tasks.
- **Automation & Agentic Tools:** Build multi-step agents and small-scale bots that connect models with data sources like SQL databases or internal apps, enabling natural language querying and process automation.
- **Collaboration & Mentorship:** Work closely with engineers and researchers to share best practices, review prompts and code, and contribute to continuous improvement of AI workflows and product integrations.

Data Analyst / Data Engineer – CellCore Biosciences (Jan 2025 – July 2025)

- **Data Architecture:** Designed and deployed a Microsoft Fabric Lakehouse ingesting 100k monthly events from Shopify, Recharge, finance, and fulfillment systems – cut reporting latency and enabled near real-time executive analytics.
- **ETL & Optimization:** Built scalable PySpark + SQL ETL pipelines consolidating 20 + sources; reduced prep time from 3 days to 45 minutes and cut manual work by 95%; optimized Spark compute and autoscaling for ~30% lower costs and doubled nightly throughput.
- **Analytics Products:** Delivered 30+ Power BI dashboards (AOV, cohort retention, funnels, inventory velocity, adoption metrics) that became the single source of truth across teams.
- **Experimentation & Modeling:** Applied Python-based ML (PrefixSpan, K-Means, Markov Chains) for product recommendation and customer segmentation; productionized a recommendation service that re-engaged ~12K dormant customers and drove \$1.3 M incremental revenue in 3 months; ran controlled bundling tests demonstrating a 33% AOV lift.
- **Data Lifecycle Ownership:** Automated KPI delivery from Salesforce via CDC pipelines + Azure Functions; created reusable SQL (CTEs, window functions, sub-queries) for segmentation and lifecycle tracking; maintained GitHub-based version control for ETL and SQL.
- **Documentation & Enablement:** Authored a 25-page data dictionary and led cross-departmental workshops that reduced ad-hoc reporting requests by 60%+.
- **M&A Integration:** Led data migration and unification for the Biocidin Botanicals acquisition—normalized 500K+ historical orders/SKUs/customers, harmonized schemas, resolved structural gaps, and delivered a unified post-acquisition analytics environment.

Data Science Trainee – Palm Beach Accountable Care Organization (PBACO) (2024)

- Developed and delivered daily, weekly, and monthly reports using **SQL Server** and **Excel** to support executive decision-making.
- Created interactive graphs, charts, and reports providing actionable insights to the Chief Information Officer and department managers.
- Automated reporting processes using **Power Automate** and ensured timely dissemination of reports to executives and key stakeholders.
- Conducted natural language processing (**NLP**) for sentiment analysis of customer reviews, providing actionable insights and recommendations to the executive team.

Data Engineering Intern – Albury Brother’s Boats (2023)

- Design, developed, and deployed scalable data pipelines: Implemented ETL processes using **Python** and **SQL** to handle large volumes of data from various sources, ensuring smooth data flow for marine engineering projects.
- Implemented data quality checks and validation procedures: Used **Python** and **SQL** to ensure the integrity and accuracy of data throughout the pipeline.
- Optimized data processing: Implemented efficient **ETL** pipelines using **Apache Spark** and automated workflows with **Apache Airflow** to improve data processing efficiency

Biotechnology Tutor – Palm Beach State College (2022)

- Tutored students in **bioinformatics**, focusing on the analysis and interpretation of complex **genetic datasets**.
- Assisted students in understanding and implementing **genetic algorithms** and **machine learning** techniques for biological data.
- Developed one-on-one and group tutoring sessions to enhance students’ proficiency in **data-driven research methods** and **biostatistics**.

PROJECTS

FISHeye – Florida Atlantic University WAVE Competition (2024)

- Created a mobile genetic detection platform prototype leveraging machine learning to identify genetic sequences with a smartphone.
- Integrated the **k-Nearest Neighbors (k-NN)** algorithms for image recognition and to enhance detection accuracy with Fluorescent In Situ Hybridization (**FISH**) and smartphone-based visualization.

Biotechnology Poster Symposium Winner – PBSC (2023)

- Conducted research on restriction endonucleases and their application in genetic research.
- Presented findings to a panel of scientists at UF Scripps, showcasing molecular biology expertise and analytical precision.

EDUCATION

Master of Science in Artificial Intelligence: Current

Bachelor of Applied Science in Information Management: 2024

Concentration: Database Administration & Data Analytics

Associate in Arts: 2023

Concentration: STEM

College Credit Certificate, Biotechnology Laboratory Specialist: 2023